

MORRIS SARDO

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EDUCATION

Royal Holloway <u>University</u> of London MSci Computer Science (Artificial Intelligence) Expected grade 2:1	Egham, Surrey September 2022 - June 2026
City and Islington College Level 3 Access to Higher Educational Diploma Engineering CGPA: 139/145	London, UK September 2021 - June 2022
City and Islington College GCSE Maths CGPA: 7	London UK September 2020 - June 2021

SKILLS

Certificate: MATLAB Onramp.

Programming Languages: Java, Python, JDBC, JavaScript, HTML, CSS, C, SQL.

Good knowledge of scikit-learn, TensorFlow, Keras, PyTorch, JUnit, Java Doc, implementing using (TTD) Test Driver Development and (VCS) Version Control System.

Machine Learning and Deep Learning:

- Experience in implementing NN in classification and regression problems.
- Experience in implementing CNN and Full Dense Layers NN.
- Experience how to detect and reduce or eliminate Overfitting as well as Underfitting.
- Knowledge on RNN.
- Knowledge in Siamese NN.
- Knowledge of implementing Transfer Learning NN.
- Knowledge of GAN and how to implement its NN.
- Knowledge of Long-Short-Term-Memory.
- Knowledge in Reinforcement Learning.

Personal interest:

I am interested in research in classification problems.

In particular:

- Detect Deepfake
- Detect Cancer. (this is still at the beginning stage).

I am interested in research in Long Short-Term Memory and how to apply it.

I am interested in research into regression problems.

In particular

- Stock prediction market (on-line Machine Learning)
- Predict weather.

CARRER HISTORY

United Kingdom, Egham Surrey.

June 2024 – September 2024

Research Placement in Royal Holloway University of London.

Summary:

Collaborating with a research team interested in the visualization of datasets from high dimensional

- space to two-dimensional space without losing the reliability of the data.

My main duties are:

- Understanding different methods of linear and nonlinear dimensional reduction like PCA, t-SNE, ISOMAP, and SOMAM, and how to plot them.
- Understanding and applying algorithms (Python) for dimensionality reduction (DR).
- Testing the reliability of the algorithms.
- Using SVC.
- Having a scrum meeting once per week with my tutor to discuss concerns, progress, and next steps.
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SUMMARY

In my first year in the Royal Holloway University of London, I was part of a team that implemented a self-driving car project. We used LEJOS Java for the mechanical aspects and OpenCV Java for the recognition part. Our team also utilized GitLab for version control, enhancing our collaborative efficiency.

During second year, I have undertaken several projects:

- Developed a calculator app capable of interpreting expressions in both standard and reverse Polish notation, implementing TDD, VCS, and the MVC design pattern. This project was particularly successful, teaching me valuable skills for professional teamwork.
- Created a database for an airport, requiring knowledge of SQL and utilizing Java and JDBC in a Java environment. This project further developed my technical and organizational skills.
- Developed a game in C that simulates dynamic memory allocation, challenging my understanding of lower-level programming concepts.
- Currently, I am part of an eight-member team developing a restaurant management system. This project uses JavaFX and CSS for the GUI, Java and JDBC for the controller and model, and adheres to TDD principles. We use GitLab for version control to maintain connectivity and collaboration.
- I am collaborating on a project which aims to implement algorithms to bring data from multidimensional spaces to two-dimensional space without losing precision.

During my current year (third year) I have undertaken several projects:

- Developed couple classification problem in Machine Learning. Those projects tested my understanding and how to apply: the conformal and non-conformal prediction, cross-conformal prediction and evaluate the performance, efficiency, so on and so for of the model.
- Developed one CNN (Convolution Neuron Net) and Full Dence Layer for classification problem. This project aimed to test my ability to implement a different model, create a state Overfitting and finally implemented a better model such that reduced or eliminated the Overfitting.
- I have implemented Black-Scholes formula, Binomial Tree methods and Monte Carlo methods algorithms to calculate the European and American call and put option.

In addition, I have developed strong skills in Optimization Problems like: Integer Programming Modelling and Branch and Bound Method, LP duality, Two-Phase Methods, Polynomial-solvable Problems, Combinatorial Optimization Problems and Construction Heuristic, Linear Programming and Simplex Method. Moreover, I developed strong knowledge in Differentiation and Derivative.

References: Available on request.